to using the subgroup's naming conventions. The rate of change depended on group size and relative size of the subgroup; see Github–economics/Centola-Becker.R.

Discoveries and inventions are often made by individuals, and it might be expected that larger populations will contain more tools and artefacts, and have a more complex cultural repertoire than smaller populations.[1112] The evidence is mixed, with population size having a positive correlation with tool complexity in some cases.[385]

Useful new knowledge is not always uniformly diffused out into the world, to be used by others; a constantly changing environment introduces uncertainty[1527] (i.e., noise is added to the knowledge signal), and influential figures may suppress use of the ideas (e.g., some physicists suppressed the use of Feynman diagrams[190]).

Conformist transmission is the hypothesis that individuals possess a propensity to preferentially adopt the cultural traits that are most frequent in the population. Under conformist transmission, the frequency of a trait among the individuals within the population provides information about the trait's adaptiveness. This psychological bias makes individuals more likely to adopt the more common traits than they would under unbiased cultural transmission. Unbiased transmission may be conceptualized in several ways: for example, if an individual copies a randomly selected individual from the population, then the transmission is unbiased; if individuals copy their parents or just their mother, then transmission is unbiased.

The rate of change in the popularity list of baby names, dog breeds and pop music can be fitted[176] by a model where most of the population, $N$, randomly copy an item on the list, containing $L$ items, and a fraction, $\mu$, select an item not currently on the list. Empirically, the turnover of items on the list has been found to be proportional to: $L\sqrt{\mu}$; a theoretical analysis[556] finds that population size does have some impact, i.e., $L\sqrt{\mu \log \frac{N}{L}}$.

Studies have found that subjects are able to socially learn agreed conventions when communicating within networks having various topologies (containing up to 48 members; the maximum experimental condition).[314] One advantage of agreed conventions is a reduction in communications effort when performing a joint task, e.g., a reduction in the number of words used.[370]

The impact of the transmission of information about new products is discussed in section 3.6.3.

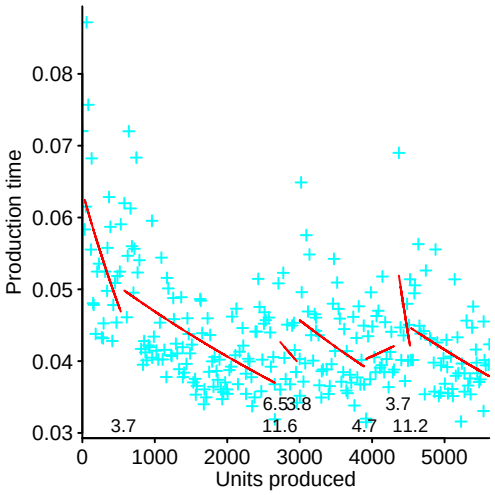



Figure 3.20: Hours required to build a car radio after the production of a given number of radios, with break periods (shown in days above x-axis); lines are regression models fitted to each production period. Data extracted from Nembhard et al.[1365] Github–Local

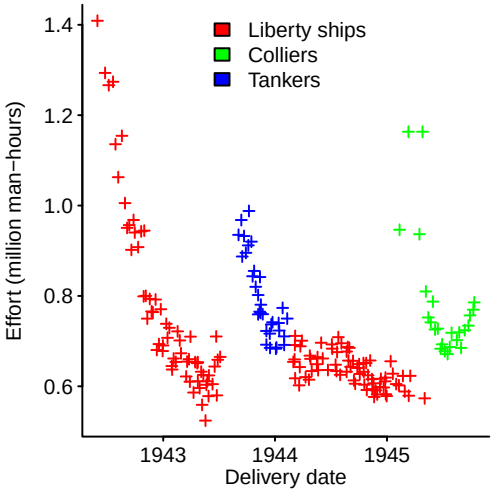



Figure 3.21: Man-hours required to build a particular kind of ship, at the Delta Shipbuilding yard, delivered on a given date (x-axis). Data from Thompson.[1833] Github–Local

## 3.4.5 Group learning and forgetting

The performance of groups, like that of individuals (see fig 2.33), improves with practice; groups also forget (in the sense that performance on a previously learned task degrades with time).[905] Industrial studies have focused on learning by doing,[1833] that is the passive learning that occurs when the same, or very similar, product is produced over time.

The impact of organizational learning during the production of multiple, identical units (e.g., a missile or airplane) can be modeled to provide a means of estimating the likely cost and timescale of producing more of the same units.[698] Various models of organizational production[868, 1248] based on connected components, where people are able to find connections between nodes that they can change to improve production, produce the power laws of organizational learning encountered in practice.

There are trade-offs to be made in investment in team learning; there can be improvements in performance and team performance can be compromised.[279]

A study by Nembhard and Osothsilp[1365] investigated the impact of learning and forgetting on the production time of car radios; various combined learning-forgetting models have been proposed, and the quality of fit of these models to the data was compared. Figure 3.20 shows the time taken to build a car radio against cumulative production, with an exponential curve fitted to each period of production (break duration in days appears above the x-axis). Note, build time increases after a break, and both a power law and exponential have been proposed as models of the forgetting process.

Writing source code differs from building hardware in that software is easily duplicated, continual reimplementation only occurs in experiments (see fig 2.36). Repetitive activities do occur when writing code, but the repetition is drawn from a wide range of activities sequenced together to create distinct functionality (patterns of code use are discussed in section 7.3).